
SOFTWARE REQUIREMENT SPECIFICATION

for
Retail Store Management System iOS
Application

Version History

Name	Date	Version
Prasad B S	03/03/2026	1.0

1. INTRODUCTION

1.1. APP VISION

This document specifies the software requirements for a **Retail Store Management System (RSMS)** tailored to a multi-country luxury goods retail chain (e.g., jewelry, watches, leather goods, couture). The RSMS will unify store operations, clienteling, product lifecycle, omnichannel fulfilment, and after-sales services while preserving the brand's high-touch experience and compliance with global retail and payments regulations.

1.2. PURPOSE OF THE APP

This project is designed as a configurable product that can be adapted and customized for various luxury retail formats, store sizes, and regional compliance needs. The primary objectives are to:

- Provide a centralized platform for product master data, pricing, inventory, and allocations across boutiques and online.
- Enable clienteling (client profiles, preferences, purchase history, wishlists) and VIP appointment management to deliver a personalized client experience.
- Support retail POS (including split tenders, tax-free, gift receipts, exchanges, returns) with offline resilience and PCI-DSS compliant payments.
- Facilitate omnichannel operations: BOPIS, BORIS, endless aisle, ship-from-store, client reservations, and remote selling.
- Implement RFID-enabled stock management for real-time visibility, cycle counts, and shrink control.
- Orchestrate after-sales services: repairs, services, authentication, warranty, and valuations.
- Offer analytics and compliance reporting for sales, inventory, client KPIs, and regulatory needs.
- Provide AI-assisted recommendations (cross-sell/up-sell), client segmentation, and demand forecasting.

Target audience: Corporate retail teams, boutique managers, sales associates, inventory controllers, merchandisers, after-sales staff, finance, and IT operations.

2. SCOPE OF THE PROJECT

Definitions:

SKU: Stock Keeping Unit; item-level product identifier.

RFID: Radio Frequency Identification tag for item-level tracking.

Clienteling: Sales approach centered on personal client relationships supported by data.

BOPIS/BORIS: Buy Online Pick-up In Store / Buy Online Return In Store.

Endless Aisle: Selling inventory not physically available in the store via other locations/DC.

AST: After-sales ticket (repairs/services/authentications).

POS: Point of Sale system.

OMS: Order Management System for omnichannel orchestration.

2.1. HIGH LEVEL FEATURES

(Add Min 5 features per user type identified) Sample features shall be as below

2.1.1.Store/Boutique Admin Management:

2.1.1.1.Store Dashboard: Sales targets vs actuals, appointments, staff shifts, stock alerts.

2.1.1.2.Staff & Shift Planning: Sales goals, commissions, performance tracking.

2.1.1.3. Local Assortment & Visual Merchandising: Planograms, lookbooks, compliance checks.

2.1.1.4.Inventory Operations: Receiving, put-away, transfers, cycle counts, discrepancy handling (RFID/mobile).

2.1.1.5.Client Events: Trunk shows, VIP previews, invitation & RSVP management..

2.1.2.Sales Associate (Client Advisor):

2.1.2.1.Clienteling: Client profiles, preferences, sizes, anniversaries, wishlists, privacy consents.

2.1.2.2.Appointment & Remote Selling: Booking, reminders, video consults, curated carts.

2.1.2.3.Assisted Selling: Guided catalogs, look builder, AI recommendations, cross-sell/upsell.

2.1.2.4.POS & Checkout: Carting, discounts, split tenders, tax-free, gift receipts, exchanges/returns.

2.1.2.5. Omnichannel: BOPIS pickup, endless-aisle order placement, ship-from-store packing slips.

2.1.3.Inventory Controller

2.1.3.1.RFID Stock Management: Item-level scans, exception reconciliation, shrink analytics.

2.1.3.2.Transfers & Replenishment: Inter-store, DC-to-store, vendor returns; ASN matching.

2.1.3.3.Compliance & Audit: Cycle count schedules, audit trails, variance signoff.

2.1.3.4.Serialization & Certificates: Serial capture, certificates of authenticity linkage/printing.

2.1.3.5.Repairs Logistics: Intake/outbound to service centers, loaners, SLA tracking.

2.1.4.After-Sales Service (Repairs & Care)

2.1.4.1.Intake & Diagnostics: AST creation, photos, condition reports, warranty validation.

2.1.4.2.Workflow: Estimation, client approval, parts allocation, repair stages, QA, completion.

2.1.4.3.Communication: Client notifications, approvals, cost estimates, payment collection.

2.1.4.4.Warranty & Authentication: Warranty lookups, authentication requests, valuation letters.

2.1.4.5>Returns to Client: Pickup scheduling, packaging, documentation.

2.2. TYPES OF USERS

(Min 3 types of users to be identified)

1. Corporate Admin / Retail Ops: Manages product, pricing, promotions, compliance; high technical proficiency.
2. Boutique Manager: Oversees store KPIs, staff, assortment, events, inventory audits; strong retail ops knowledge.
3. Sales Associate (Client Advisor): Clienteling, selling, POS, omnichannel; strong client service skills.
4. Inventory Controller: Stock accuracy, RFID, transfers, audits; operations focus.
5. After-Sales Specialist: Repairs, warranty, authentication; technical product knowledge.

2.3. TECHNICAL SPECIFICATIONS

Devices: iPhone, iPad, OS: iOS 26+.

Frameworks: SwiftUI, Core ML, Vision, App Intents.

Architecture: MVVM with Swift Concurrency.

Security: Passkeys, encryption, role-based access, GDPR compliance.

3. NON-FUNCTIONAL REQUIREMENTS

3.1. PERFORMANCE

- Load Time: The application will be responsive and load quickly, even on slow network connections
- The application will be able to handle high volumes of data (xxx MB) and support large numbers of users (1000s).
- Zero memory leaks
- Zero constraint issues (warnings and conflict errors)

3.2. SECURITY

- The application will use secure login protocols, such as password protection and two-factor authentication, to ensure the privacy and security of user information. Should have role based authentication for accessing functional modules,
 - The application will use secure data storage and encryption to protect sensitive information from unauthorised access.
 - The application will comply with relevant data privacy regulations
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3.3. USABILITY

- The application will have a user-friendly interface that is easy to navigate and understand.

3.4. SCALABILITY

- The application should be scalable to accommodate growth and changes in user needs.
- The application should be able to support new features and functionalities as needed.

3.5. RELIABILITY

- The application should be reliable and available at all times, with minimal downtime.
- The application should be able to recover from failures and maintain data consistency in case of errors.
- Zero leaks, zero constraint warnings, zero warnings

3.6. ACCESSIBILITY

- How easily people with the widest range of capabilities can use the system.

4. DETAILED FUNCTIONAL REQUIREMENTS (SAMPLE)

4.1. Store/Boutique Admin Management:

- Display a real-time dashboard tracking sales actuals vs. targets, daily appointments, staff shifts, and stock alerts.
- Scheduling shifts, track performance goals, and dynamically calculate associate commissions.
- Upload photos to verify localized visual merchandising compliance against corporate planograms and lookbooks.
- Mobile/RFID workflows for receiving, put-away, transfers, and cycle counts, including discrepancy logging and handling.
- Creation of VIP events and trunk shows, managing digital invitations and tracking real-time RSVP statuses.

4.2. Sales Associate (Client Advisor):

- Maintain comprehensive client profiles capturing sizes, preferences, wishlists, milestone dates, and privacy consents.
 - Manage bookings, send automated reminders, conduct video consults, and share shoppable curated carts.
 - Guided digital catalogs and utilize an AI engine to recommend cross-sell and upsell items based on client history.
 - Transactions processing with support for split tenders, discounts, tax-free documentation, gift receipts, and returns.
 - Alert associates of BOPIS (Buy Online, Pick Up In-Store) orders and provide workflows for endless-aisle ordering and ship-from-store packing.
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4.3. Inventory Controller:

- Processing item-level RFID scans to instantly flag physical vs. systemic inventory exceptions and generate shrink analytics.
- Automate Advanced Shipping Notice (ASN) matching for inter-store transfers, DC-to-store replenishment, and vendor returns.
- Schedule regular cycle counts and maintain immutable audit trails requiring managerial signoff for variance discrepancies.
- Serial number capture for high-value merchandise and automatically link/print localized Certificates of Authenticity.
- Track inbound/outbound shipments to service centers, monitor SLA compliance, and manage loaner item inventory.

4.4. After-Sales Service (Repairs & Care):

- Generate After-Sales Tickets (AST) to capture preliminary item condition photos, diagnostic reports, and warranty validation.
- Track repair stages (estimation, parts allocation, completion) and mandate a Quality Assurance (QA) checklist prior to finalization.
- Automate client notifications for status updates and provide a secure digital gateway for cost estimate approvals and payment collection.
- Provide lookup tools for active warranties and facilitate official authentication requests or valuation letters from the corporate team.
- Schedule client pickups, generate return packaging documentation, and track outbound shipments for completed repairs.

5. SUBMISSIONS

5.1. Code base

5.2. App Video demo

5.3. Memory Profile Screenshot

5.4. Flow diagram
